ENCI335
Structural Analysis and Systems I
Tutorial 2 Solution

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Objectives

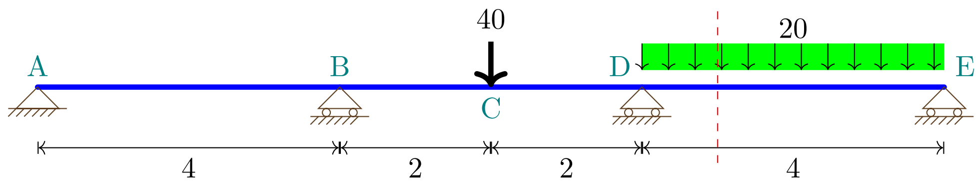
This tutorial provides an opportunity to practice the following:

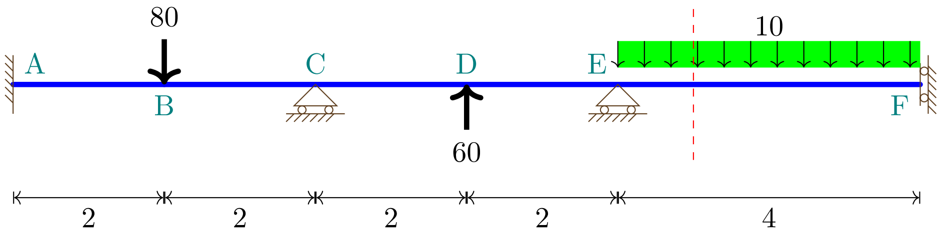
- Analysis of statically indeterminate beams
- Sketching of free-body diagrams
- Superposition and the Force Method
- Moment-area theorems

Question 1 FBD, $M(x)$, BMD and v.

Given the following beams:

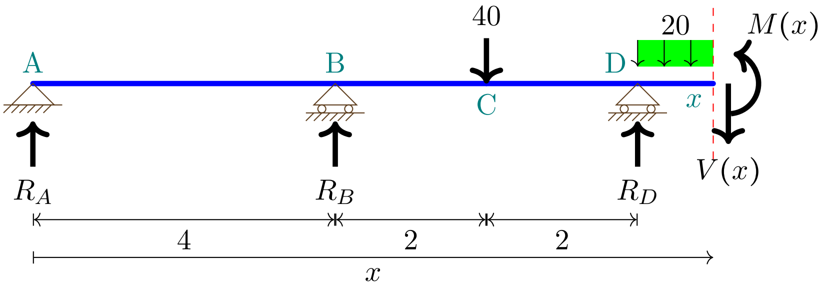
- Sketch free-body diagrams for section cuts taken at the locations approximately indicated by the red dashed lines. Start from the **left** support and label the unknown reactions as R_A, M_A, etc. Ignore horizontal components.
- Express the bending moment $M(x)$ as a general function of x (measured from the left end) for these section cuts using Macaulay brackets.
- Sketch the qualitative deflected shape, BMD, and SFD for the entire beam.





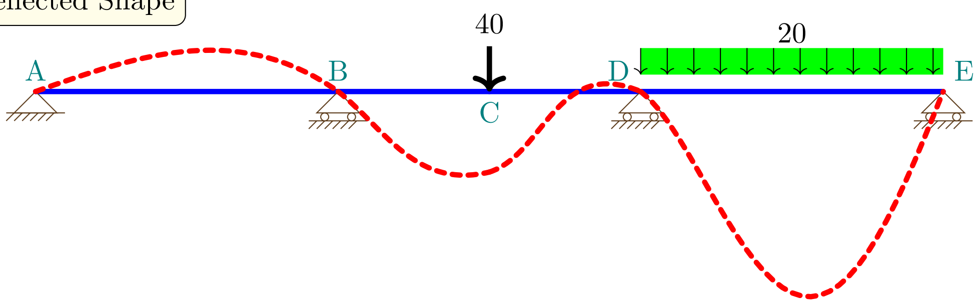
Solution:

Beam 1

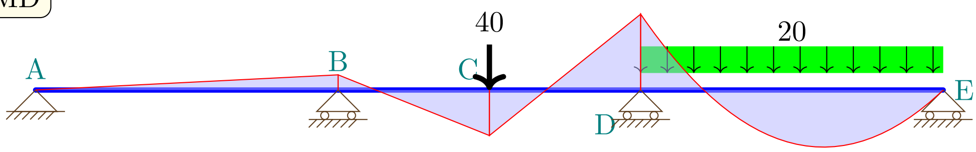


$$M(x) = R_A x + R_B\langle x-4\rangle - 40\langle x-6\rangle + R_D\langle x-8\rangle - \frac{20\langle x-8\rangle^2}{2} \tag{1}$$

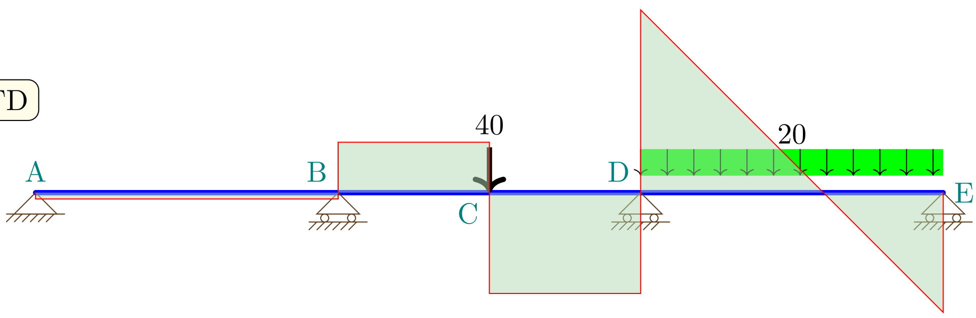
Deflected Shape



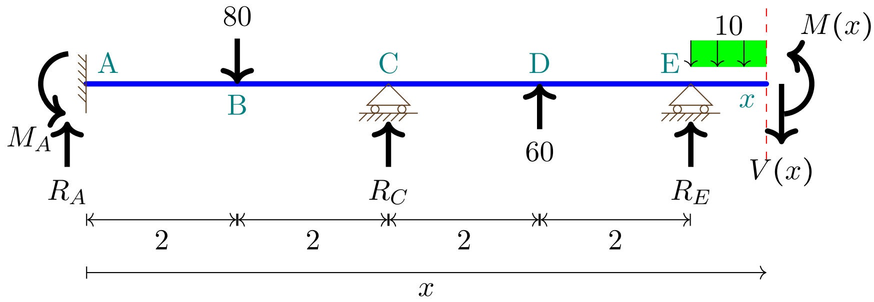
BMD



SFD

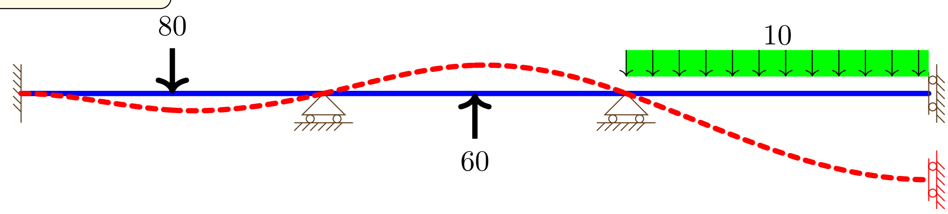


Beam 2

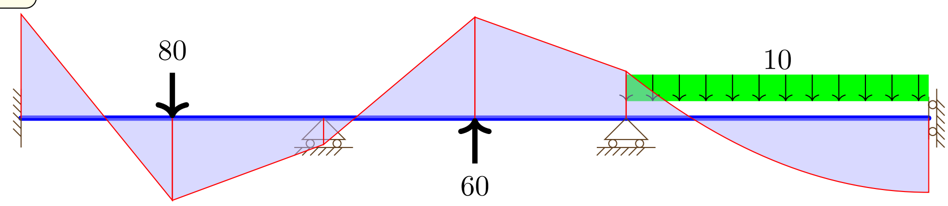


$$M(x) = -M_A + R_A x - 80\langle x-2\rangle + R_C\langle x-4\rangle + 60\langle x-6\rangle + R_E\langle x-8\rangle - \frac{10\langle x-8\rangle^2}{2} \qquad (2)$$

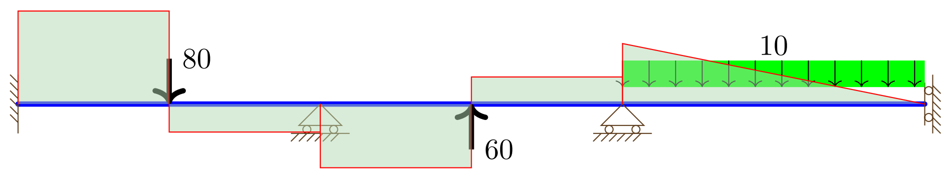
Deflected Shape



BMD

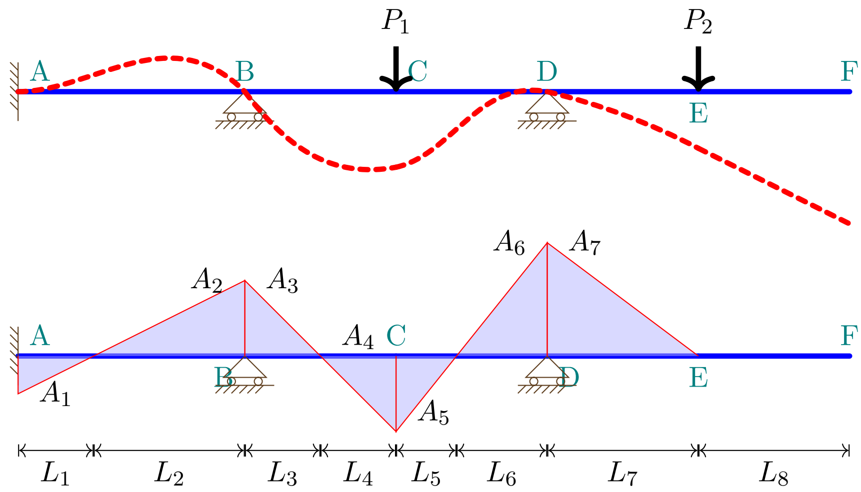


SFD



Question 2 Moment area theorems

Consider a beam with its deflected shape and **curvature** diagram shown as follows.



Use the Moment-Area Theorems to determine the following in terms of the triangle area A_i (they carry a sign) and length L_i: a) δ_{FD}, b) θ_{BA}, c) δ_{EB}, d) θ_{DB}, and e) δ_{CB}.

Solution:

$$\delta_{FD} = A_7(L_8 + 2L_7/3) \tag{3}$$

$$\theta_{BA} = A_1 + A_2 \tag{4}$$

$$\begin{aligned}\delta_{EB} &= A_7(2L_7/3) + A_6(L_7 + L_6/3) + A_5(L_7 + L_6 + 2L_5/3) \\ &\quad + A_4(L_7 + L_6 + L_5 + L_4/3) + A_3(L_7 + L_6 + L_5 + L_4 + 2L_3/3)\end{aligned} \tag{5}$$

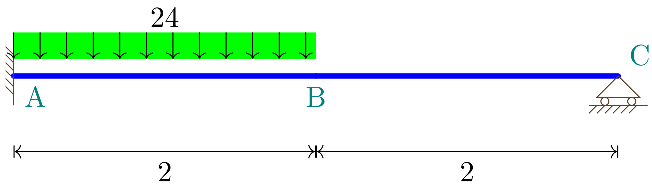
$$\theta_{DB} = A_6 + A_5 + A_4 + A_3 \tag{6}$$

$$\delta_{CB} = A_4L_4/3 + A_3(L_4 + 2L_3/3) \tag{7}$$

Please note that the values of areas A_2, A_3, A_6, and A_7 carry a negative sign due to the negative curvature. Therefore, using addition ('+') in the above equations to compute rotations and tangential deviations is correct. It is good practice to always include the appropriate signs for curvature areas (or moment areas).

Question 3 Analysis of SI beam

Solve for the support reactions, calculate the deflection and rotations at points B and C for the following beam of a uniform EI using the methods: a) the integration method, b) Superposition/force method (check my slides for the table), and c) Moment area theorems. Draw BMD, SFD and sketch the deflected shape.



Solution:

- Integration method:

$$\frac{dV}{dx} = -24 + 24\langle x-2\rangle^0 \tag{8}$$
$$\implies V(x) = -24x + 24\langle x-2\rangle^1 + C_1 \tag{9}$$
$$\implies M(x) = -12x^2 + 12\langle x-2\rangle^2 + C_1 x + C_2 \tag{10}$$
$$M(4) = 0 \implies 4C_1 + C_2 = 144 \tag{11}$$
$$EI\kappa = -12x^2 + 12\langle x-2\rangle^2 + C_1 x + C_2 \tag{12}$$
$$EI\theta = -4x^3 + 4\langle x-2\rangle^3 + \frac{C_1}{2}x^2 + C_2 x + C_3 \tag{13}$$
$$EIv = -x^4 + \langle x-2\rangle^4 + \frac{C_1}{6}x^3 + \frac{C_2}{2}x^2 + C_3 x + C_4 \tag{14}$$
$$\theta(0) = v(0) = 0 \implies C_3 = C_4 = 0 \tag{15}$$
$$v(4) = 0 \implies 4C_1 + 3C_2 = 90 \tag{16}$$

Together with $4C_1 + C_2 = 144$, we solve $C_1 = 42.75$, and $C_2 = -27$. So

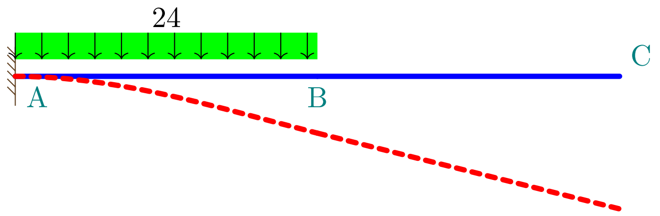
$$V(x) = -24x + 24\langle x-2\rangle^1 + 42.75 \tag{17}$$
$$M(x) = -12x^2 + 12\langle x-2\rangle^2 + 42.75x - 27 \tag{18}$$
$$EI\theta = -4x^3 + 4\langle x-2\rangle^3 + \frac{42.75}{2}x^2 - 27x \tag{19}$$
$$EIv = -x^4 + \langle x-2\rangle^4 + \frac{42.75}{6}x^3 - 13.5x^2 \tag{20}$$

We can obtain the reactions and the deflection and rotation at A, B & C as follows

$$R_A = V(0) = 42.75 \tag{21}$$
$$R_B = -V(4) = 5.25 \tag{22}$$
$$M_A = M(0) = -27 \tag{23}$$
$$\theta_B = \theta(2) = -0.5/EI \tag{24}$$
$$v_B = v(2) = -13/EI \tag{25}$$
$$\theta_C = \theta(4) = 10/EI \tag{26}$$

- Force method

  The number of reactions $n_R = 3$. The number of equilibrium equations $n_E = 2$. Therefore, the degree of static indeterminacy $SI = 1$. Set the reaction at point C as the redundant. Let's first work on the following beam with the redundant force removed. Its deflected shape is given as

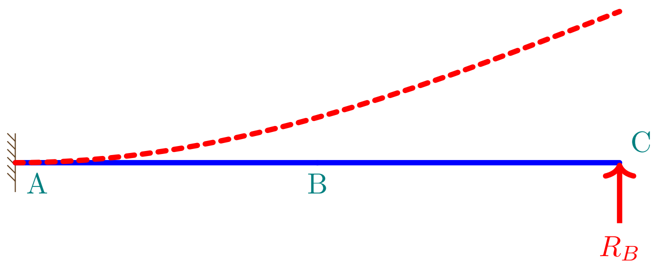
  

  Since there is no load between B and C. It implies that beam BC remains straight, and $\theta_C = \theta_B$, and $v_C = v_B + \theta_B L_{BC}$.

  From the beam deflection table, we can obtain that

  $$v_B = \frac{wL^4}{8EI} = \frac{-24 \cdot 2^4}{8EI} = \frac{-48}{EI} \tag{27}$$
  $$\theta_B = \frac{-wL^3}{6EI} = \frac{-24 \cdot 2^3}{6EI} = \frac{-32}{EI} \tag{28}$$
  $$v_C = v_B + L_{BC}\theta_B = \frac{-112}{EI} \tag{29}$$

  Now, put the redundant force R_B back:

  So, according to the table, the vertical deflection $v_C = \frac{R_B L^3}{3EI} = \frac{64R_B}{3EI}$.

  According to the compatibility condition that

  $$v_C = \frac{-112}{EI} + \frac{64R_B}{3EI} = 0 \tag{30}$$

  It implies $R_B = 5.25$, as expected.

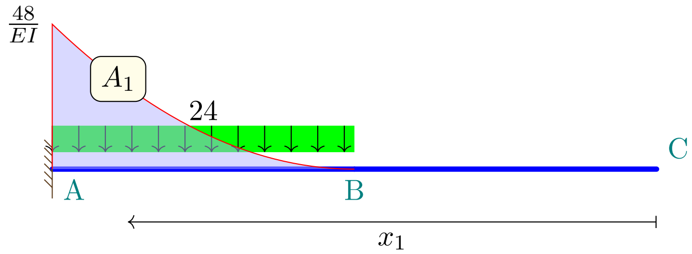
- Moment-Area theorems:

  We know the following relationship:

$$v_C = v_A + \theta_A L_{AC} + \delta_{CA} \tag{31}$$

Knowing that $v_C = v_A = \theta_A = 0$, it results in $\delta_{CA} = 0$. We need to determine what value of R_B would make $\delta_{CA} = 0$.

Let's draw the M/EI of the beam without the redundant

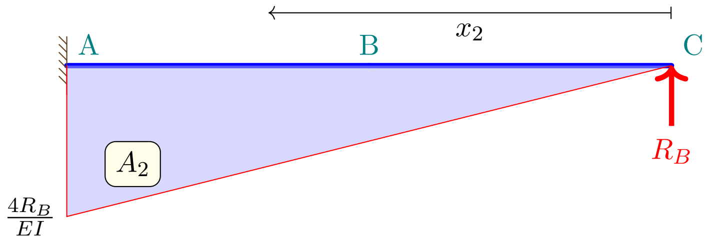


$$A_1 = \frac{-48}{EI} \times 2 \times 1/3 = \frac{-32}{EI} \tag{32}$$

$$x_1 = 3/4 \times 2 + 2 = 3.5 \tag{33}$$

$$\delta_{CA1} = A_1 \times x_1 = \frac{-112}{EI} \tag{34}$$

Let's draw the M/EI of the beam with the redundant only



$$A_2 = \frac{4R_B}{EI} \times 4 \times 1/2 = \frac{8R_B}{EI} \tag{35}$$

$$x_2 = 4 \times 2/3 = 8/3 \tag{36}$$

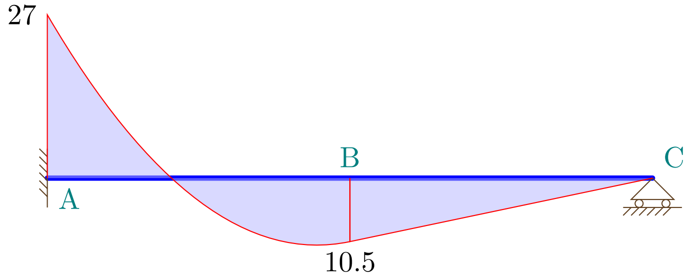
$$\delta_{CA2} = A_2 \times x_2 = \frac{64R_B}{3EI} \tag{37}$$

The compatibility condition requires

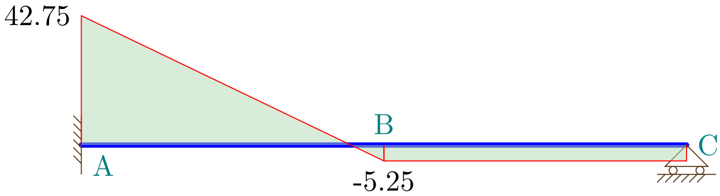
$$\delta_{CA} = \delta_{CA1} + \delta_{CA2} = \frac{-112}{EI} + \frac{64R_B}{3EI} = 0 \implies R_B = 5.25 \tag{38}$$

- BMD, SFD and Deflected Shape

BMD:



SFD:



Deflected shape: